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NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » Design and Analysis of Algorithms (course)

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● Quiz: Week 1
Quiz
(assessment?
name=234)

Week 1 Quiz

The due date for submitting this assignment has passed.

Due on 2025-08-06, 23:59 IST.

Assignment submitted on 2025-08-06, 23:30 IST

All questions carry equal weightage. You may submit as many times as you like within the deadline. Your final submission will be graded.

1) What does $f(2000, 3)$ return??

```
f(m,n) {
    ans = 1;
    count = 0;
    while (ans <= m) {
        count = count + 1;
        ans = ans * n;
    }
    return(count)
}
```

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Yes, the answer is correct.

Score: 2

Feedback:

The loop executes till we reach a power of 3 that exceeds 2000. At the while condition, we have the invariant that $ans == 3^{count}$. Since 3^6 is 729 and 3^7 is 2187, the value of count is 7 when the function returns.

Accepted Answers:

(Type: Numeric) 7



**Week 2 :
Searching
and sorting
()**

**Week 2 Quiz
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**Week 3 :
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**Week 4 :
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**Week 5: Data
Structures:
Union-Find
and Heaps ()**

**Week 5 :
Divide and
Conquer ()**

**Week 5 Quiz
()**

2) Suppose someone designs a new airline routing algorithm called MagicPath and **2 points** claims that its worst-case complexity is $O(n^2 \log n)$. Which of the following statements is inconsistent with this claim.

- ☐ For every n , for every input of size n , MagicPath is able to solve the problem in time proportional to n^2 .
- ☐ For some n , for every input of size n , MagicPath is able to solve the problem in time proportional to n^2 .
- ☐ For every sufficiently large n , there is an input of size n for which MagicPath requires time proportional to n^2 .
- ☒ For every sufficiently large n , there is an input of size n for which MagicPath requires time proportional to n^3 .

Yes, the answer is correct.

Score: 2

Feedback:

$O(n^2 \log n)$ is an upper bound on worst-case complexity. It does not force any input to actually take that much time to solve. So the only contradiction is the statement that there are infinitely many n for which there are inputs of size n that take time $O(n^3)$.

Accepted Answers:

For every sufficiently large n , there is an input of size n for which MagicPath requires time proportional to n^3 .

3) You are executing an algorithm with worst-case time complexity $O(n^4)$ on a CPU **2 points** that can perform 10^8 operations per second. Which of the following is the most accurate guarantee for the time required to solve a worst case input of size 800?

- ☐ Under 5 minutes
- ☒ Under 5 hours
- ☐ Under 5 days
- ☐ Under 5 weeks

Yes, the answer is correct.

Score: 2

Feedback:

$800^4 = 4096 \times 10^8$. 4096 seconds is about 68.3 minutes, so the best answer is under 5 hours.

Accepted Answers:

Under 5 hours

4) Suppose $f(n)$ is $n^2 \log n$. Consider the following statements. **2 points**

- (A) $f(n)$ is $O(n \sqrt{n})$
- (B) $f(n)$ is $O(n^2 \sqrt{n})$
- (C) $f(n)$ is $O(n^3)$

Which of the following is true?

- ☐ (A), (B) and (C) are all not true.
- ☒ (B) and (C) are true but (A) is not true.
- ☐ (B) is true but (A) and (C) are not true.
- ☐ (A) and (B) are true but (C) is not true.

Yes, the answer is correct.

Week 6: Data Structures: Search Trees ()

Week 6: Greedy Algorithms ()

Week 6 Quiz ()

Week 6 Programming Assignment ()

Week 7: Dynamic Programming ()

Week 7 Quiz ()

Week 7 Programming Assignment ()

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Score: 2

Feedback:

$n^2 \log n$ is not $O(n \sqrt{n})$ so (A) is false. Both (B) and (C) are true.

Accepted Answers:

(B) and (C) are true but (A) is not true.

5) In the code fragment below, first and last are integer values and composite(x) is a function that returns true if x is not a prime number and false otherwise.

2 points

```
i = 0; j = 0; k = 0;
for (m = last; m >= first; m = m - 1){
    k = k + m;
    if (composite(m)){
        i = i + m;
    }else{
        j = j - m;
    }
}

if (...) {
    print("True");
}else{
    print("False");
}
```

Which of the following expressions can we put in place of the missing if condition (...) to ensure that the program prints "True"?

- ☐ $k == i + j$
- ☒ $k == i - j$
- ☐ $k == j - i$
- ☐ None of the other options is universally true. The expression depends on the values of first and last.

Yes, the answer is correct.

Score: 2

Feedback:

In every iteration, the increase in k is matched by an increase in i or a decrease in j. Hence, $k = i - j$ is an invariant. If m is composite, this changes to $(k+m) = (i+m) - j$. If m is not composite, this changes to $(k+m) = i - (j - m)$.

Accepted Answers:

$k == i - j$